Project: Worked examples
Title: Column buckling (Perry–Robertson)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Compressive strength and resistance of a pin-ended strut of a given effective length, from Euler's critical stress reduced for initial imperfection by the Perry–Robertson formula, with the Robertson constant chosen by strut curve; utilisation against an applied axial load.

WHAT IT CALCULATES	RELATION
Radius of gyration and slenderness	$r = \sqrt{(I/A)}$; $\lambda = L_E/r$
Euler buckling stress and load	$p_E = \pi^2 E/\lambda^2$; $P_E = p_E \cdot A$
Limiting slenderness, below which the strut is not reduced for buckling	$\lambda_0 = 0.2 \cdot \sqrt{(\pi^2 E/p_y)}$
Perry factor (Robertson constant a = 2.0, 3.5, 5.5, 8.0 for curves a–d)	$\eta = \max(0.001 \cdot a \cdot (\lambda - \lambda_0), 0)$
Compressive strength by the Perry–Robertson formula	$\varphi = (p_y + (\eta + 1) \cdot p_E)/2$; $p_c = p_E \cdot p_y/(\varphi + \sqrt{(\varphi^2 - p_E \cdot p_y)})$
Compression resistance and utilisation	$P_c = p_c \cdot A$; $u = P/P_c$

Basis: Compressive strength (Perry strut formula)[1](Annex C); Maximum slenderness[1](4.7.3.2); Allocation of strut curve[1](Table 23).

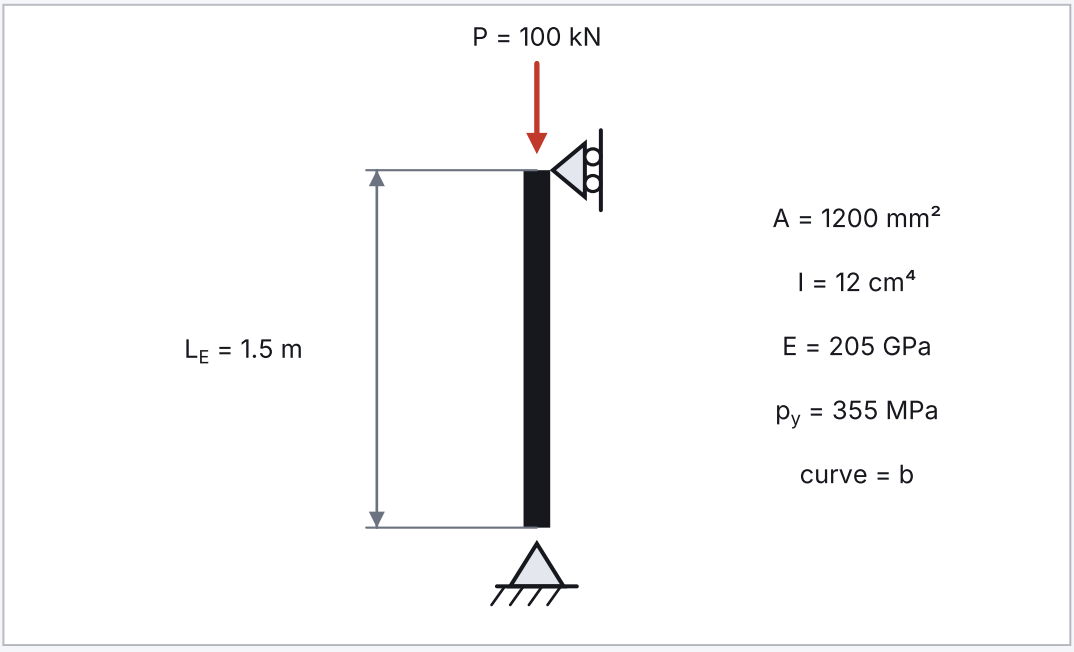


Figure 1 — A pin-ended strut under axial compression, effective length L_E. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_E	Effective length	1.5	m	drawing: DRG-ST-411 rev A	0.1 … 50 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Cross-section area	1200	mm^2	spec: SPEC-ST-02 section schedule	10 … 1000000 mm^2
I	Second moment of area about the weaker axis	12	cm^4	spec: SPEC-ST-02 section schedule	0.01 … 10000000 cm^4
E	Modulus of elasticity	205	GPa	spec: BS 5950-1 3.1.3: E = 205 GPa	50 … 450 GPa
p_y	Design strength (yield)	355	MPa	spec: SPEC-ST-02: S355	100 … 1000 MPa
curve	Strut curve (selects the Robertson constant)	b		spec: BS 5950-1 Table 23: hot-finished tube, curve b	any
P	Applied axial load	100	kN	assumption: Factored brace force	0 … 100000 kN

3. Assumptions

A1. Pin-ended, prismatic strut under axial load only; buckling about the axis of the given second moment. (Effective length carries the end restraint.)
A2. Initial imperfection represented by the Perry factor $\eta = 0.001 \cdot a \cdot (\lambda - \lambda_0)$, with a from the strut curve. (Robertson's fit to strut tests.)
A3. P (Applied axial load) is taken as 100 kN. (Factored brace force)

4. Working

[8] $a_R = 3.5$ [1](Annex C)
= 3.5
Robertson constant for strut curve b
[9] r = sqrt(I / A)
= sqrt(12 cm⁴ / 1200 mm²)
= 10 mm
Radius of gyration
[10] $\lambda = L_E / r$
= 1.5 m / 10 mm
= 150
Slenderness
[11] p_E = pi^2 * E / λ^2 [1](Annex C)
= π^2 × 205 GPa / 150^2
= 89.923 MPa
Euler critical stress
[12] $P_E = p_E * A$
= 89.923 MPa × 1200 mm²
= 107.91 kN
Euler critical load
[13] λ_0 = 0.2 * sqrt(pi^2 * E / p_y) [1](Annex C)
= 0.2 × sqrt(π^2 × 205 GPa / 355 MPa)
= 15.099
Limiting slenderness
[14] η = max(0.001 * a_R * (λ - λ_0), 0) [1](Annex C)
= max(0.001 × 3.5 × (150 − 15.099), 0)
= 0.47215
Perry factor (initial imperfection)

[15] $\varphi = (p_y + (\eta + 1) * p_E) / 2$
$= (355\ \text{MPa} + (0.47215 + 1) \times 89.923\ \text{MPa}) / 2$
$= 243.69\ \text{MPa}$
Perry–Robertson φ

[16] $p_c = p_E * p_y / (\varphi + \text{sqrt}(\varphi^2 - p_E * p_y))$ [1](Annex C)
$= 89.923\ \text{MPa} \times 355\ \text{MPa} / (243.69\ \text{MPa} + \text{sqrt}((243.69\ \text{MPa})^2 - 89.923\ \text{MPa} \times 355\ \text{MPa}))$
$= 77.973\ \text{MPa}$
Compressive strength

[17] $P_c = p_c * A$
$= 77.973\ \text{MPa} \times 1200\ \text{mm}^2$
$= 93.567\ \text{kN}$
Compression resistance

[18] $u = P / P_c$
$= 100\ \text{kN} / 93.567\ \text{kN}$
$= 1.0687$
Utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
r	10.0 mm	Radius of gyration
λ	150	Slenderness
p_E	89.9 MPa	Euler stress
P_E	107.91 kN	Euler load
λ_0	15.099	Limiting slenderness
η	0.47215	Perry factor
p_c	78.0 MPa	Compressive strength
P_c	93.57 kN	Compression resistance
u	1.07	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Axial load against compression resistance	u = 1.07	u ≤ 1	1.07	[1](Annex C)

7. Applicability

None.

8. Verification

Template struct.column-buckling@1.0.1 Content hash b92d254599e2302a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Structural Use of Steelwork in Building—Part 1: Code of Practice for Design—Rolled and Welded Sections. BS 5950-1:2000. BSI; 2000.